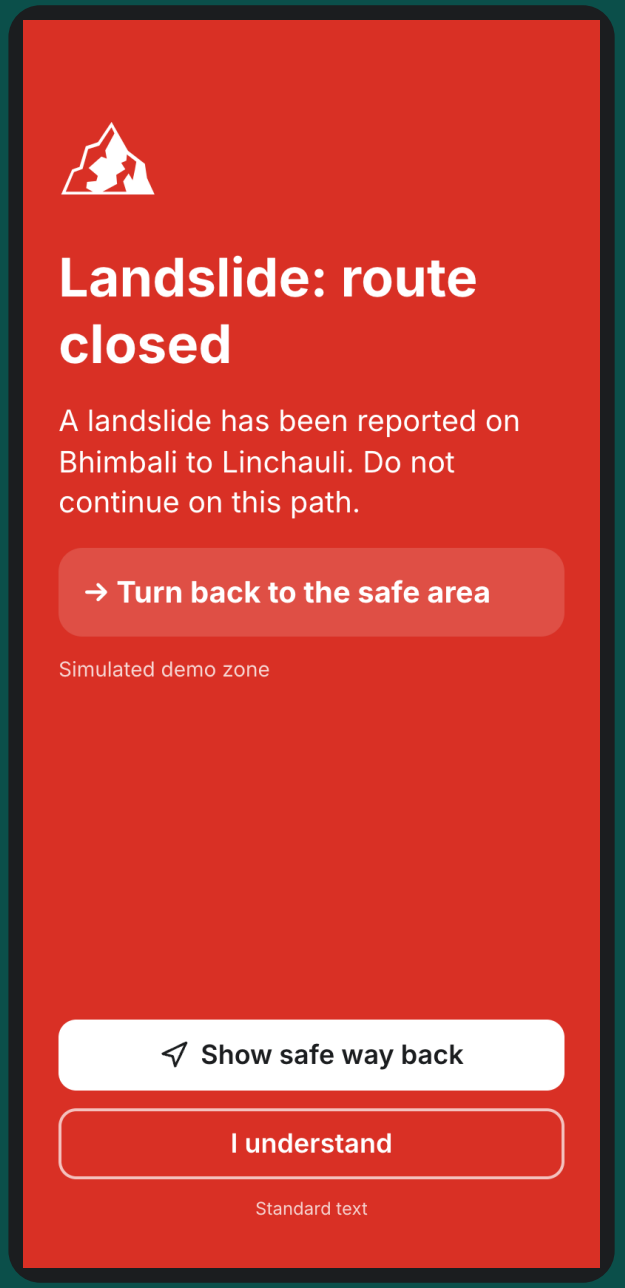
SafeTrail

Warn early. Never lose an SOS. Prove every response.

BIOS V2, Thapar Institute, Patiala

Prune challenge: Ship It Lean, AI Apps on a Budget

[TO FILL: team name and members]



Three things fail at the same moment

In October 2023, a glacial lake burst in North Sikkim. Floods cut off the roads to Lachen and Lachung, and thousands of tourists were stranded. Every season, pilgrims on the Kedarnath route face landslides and crushing crowds.

1

Nobody warns you

The route is closed or the crowd ahead is dangerous, and you find out too late.

2

There is no signal

Long stretches of these routes have no mobile data when you need it most.

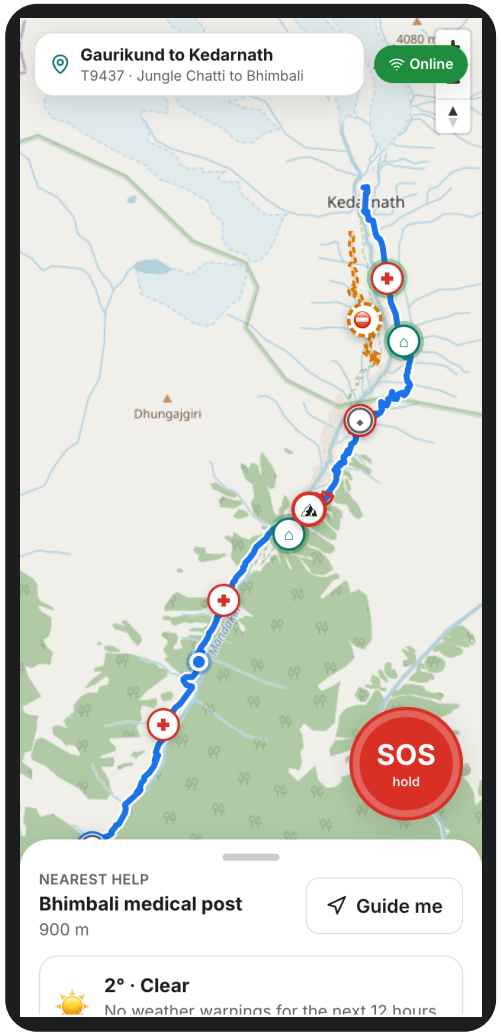
3

No shared language

You can't explain what happened to the people who can help.

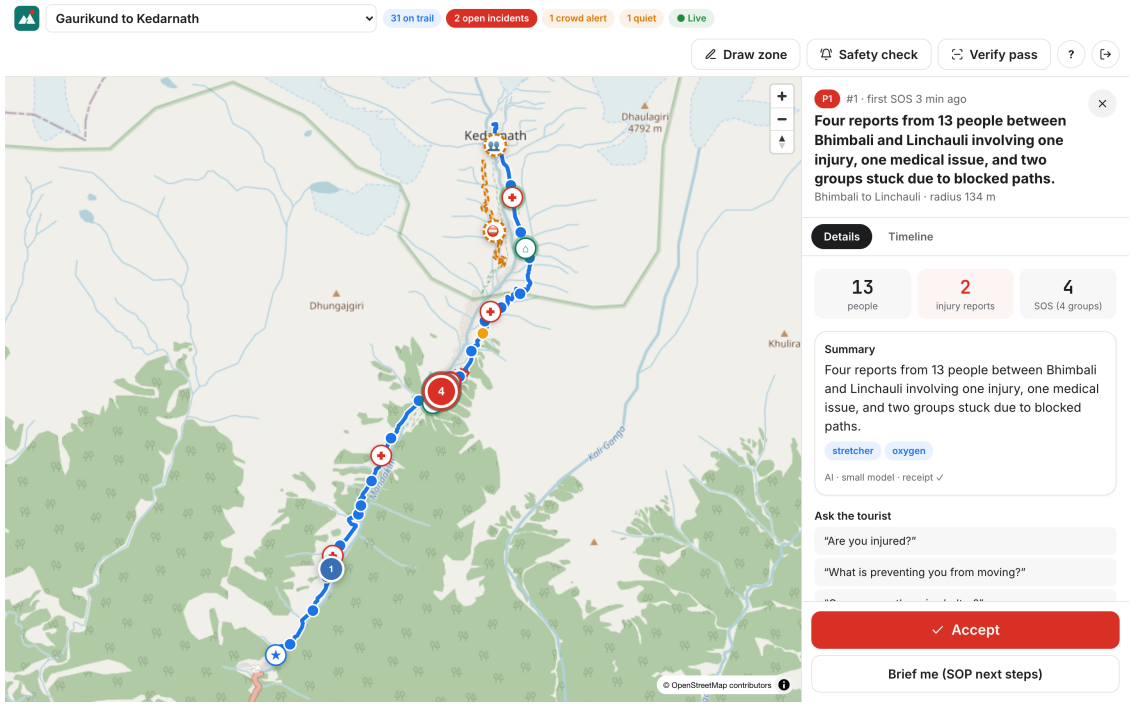
Responders get late, vague, duplicate calls.

One app for the tourist and the responder



Tourist

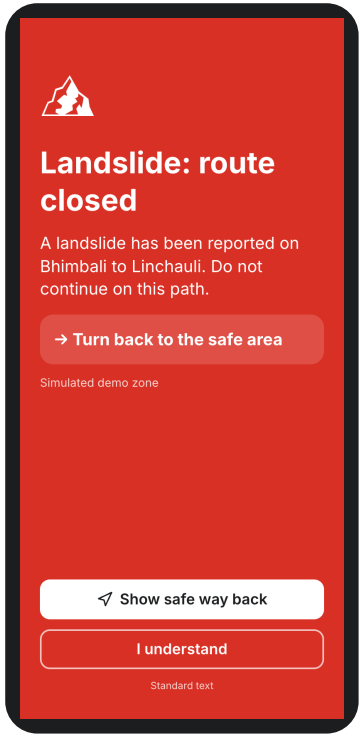
- Danger zone and crowd warnings
- Offline safety pack
- One-hold SOS
- "Are you safe?" check-in
- Family link, status only
- Signed pass



Responder

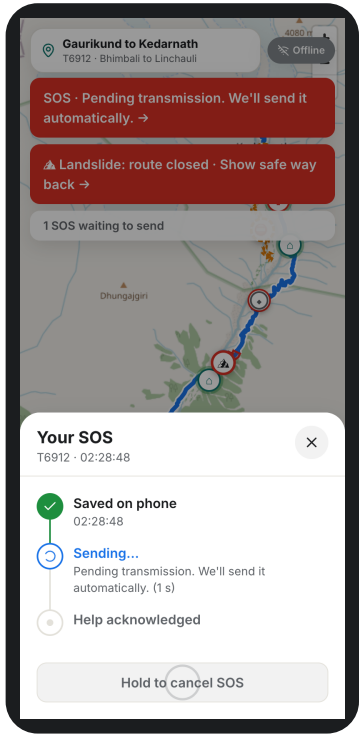
- Live map and drawn danger zones
- One grouped incident, exact counts
- Reports in their language
- Next steps, then Accept

From danger to proof in five steps



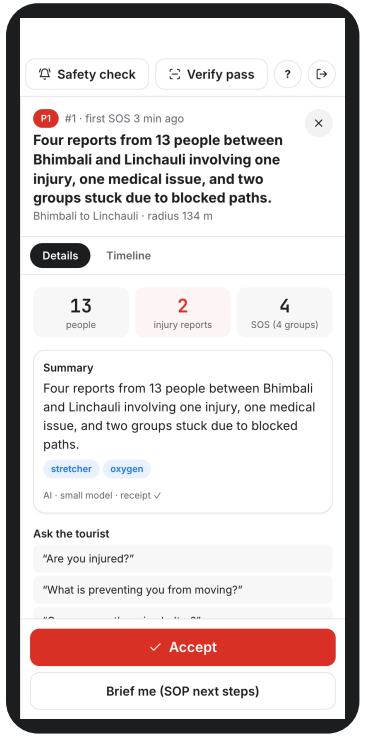
1 Warn

Alerts come from the phone, even offline



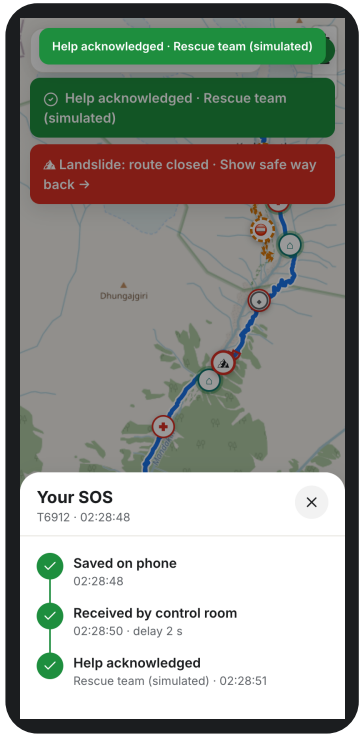
2 Save

SOS is saved at once, sent on first signal



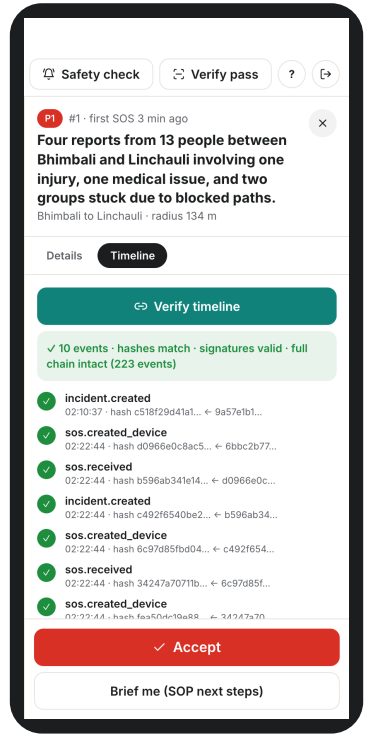
3 Group

150 m, 30 minutes: one incident, exact counts



4 Accept

The tourist sees "Help acknowledged"



5 Prove

A signed log: change one line and it breaks

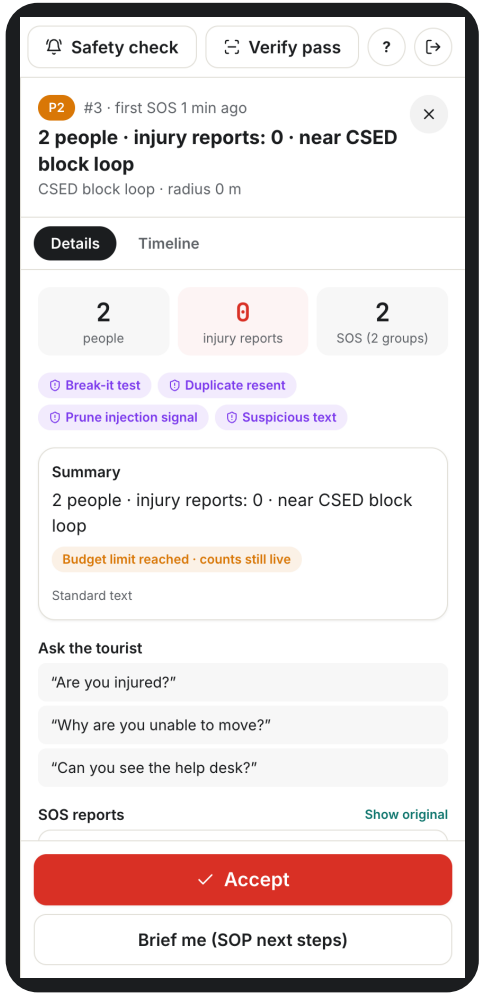
The AI is never in the emergency path

Plain code, works with the AI off

- Zone alerts and crowd detection
- SOS saving and sending
- Grouping and counts
- Check-ins and pass checks
- Signed log and family link

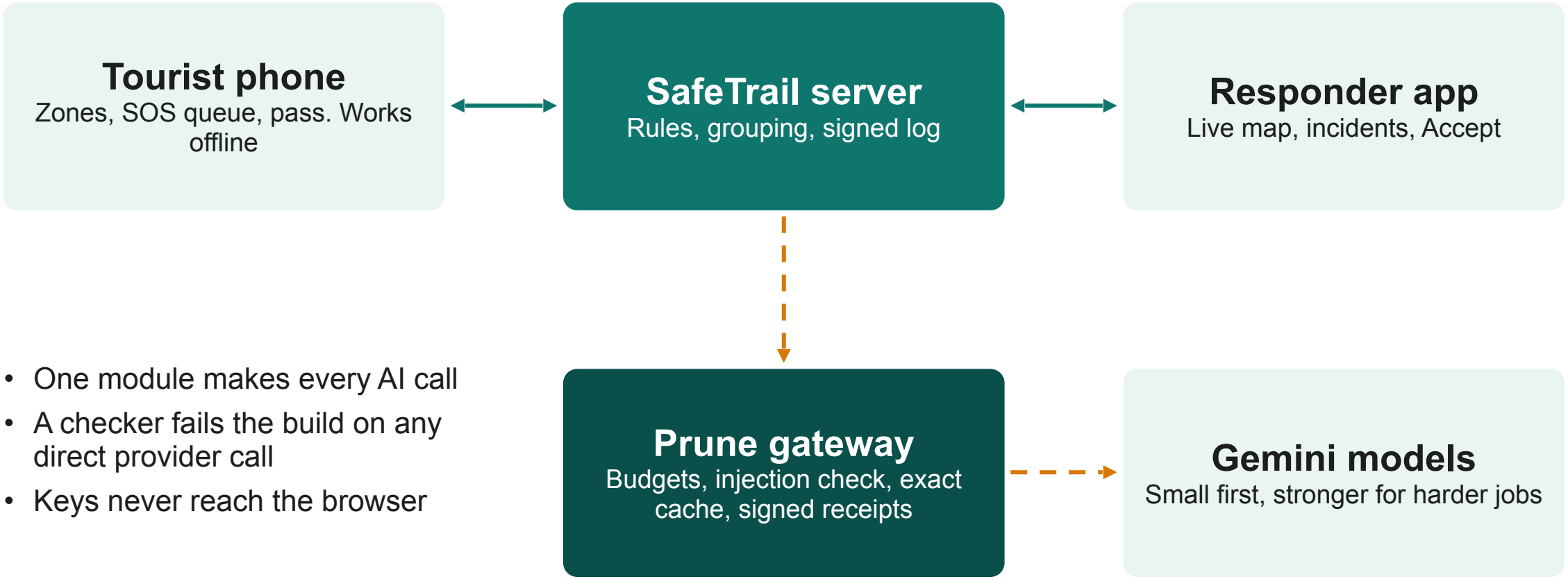
AI writes and translates only

- Safety packs
- Zone warnings
- Incident reports and summaries
- Responder next steps
- SMS details and permit cards



Rules decide. If the AI is slow, down or over budget, standard text shows, and it appears first anyway.

Every AI call goes through one gate

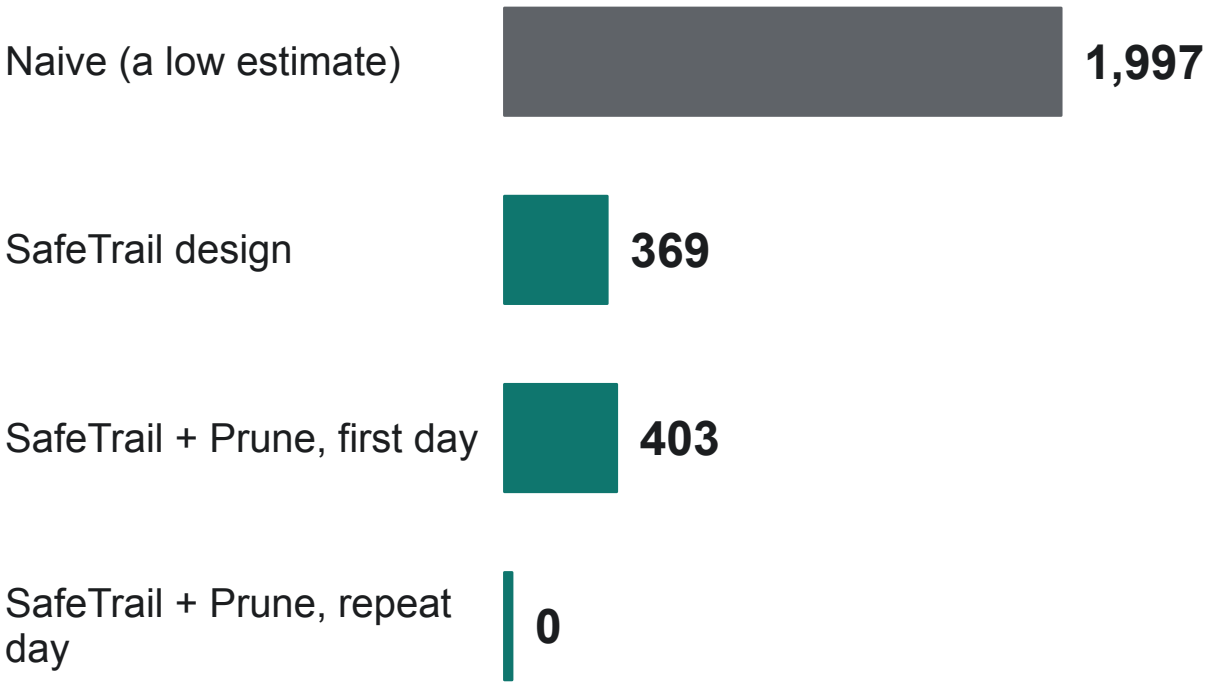


- One module makes every AI call
- A checker fails the build on any direct provider call
- Keys never reach the browser

Dotted lines are the AI. Everything on the solid lines works without it.

We spend AI only where words are needed

Tokens per monitored trip



3 paise

AI cost per monitored trip

10 paise

AI cost per emergency

82%

fewer tokens per trip than the naive way

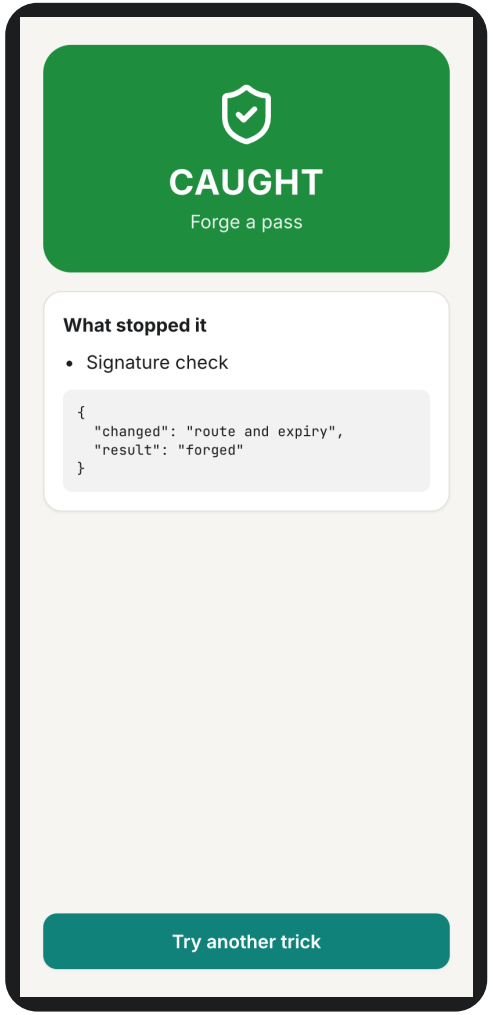
72%

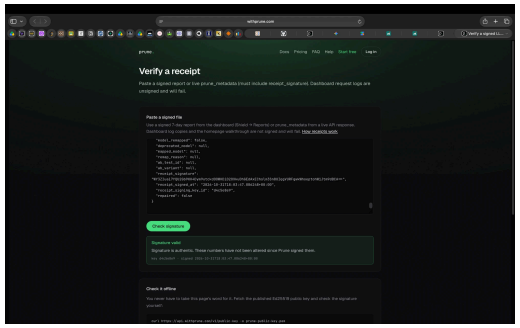
of repeat-day calls served from cache

100 simulated trips on the small model. Cost is Google's token counts times Google's list price. The live demo runs on a free key.

Built to fail safely, and tested on real prompts

- **Validators**
  Invented numbers or places fall back to standard text
- **Privacy**
  No names, phone numbers or ID numbers go into a prompt
- **Budgets**
  Prune stops a job over budget, and the app carries on
- **Injection**
  English tricks in an SOS note get flagged, and no SOS is blocked
- **Receipts**
  Signed by Prune and verified on its own site
- **Our best catch**
  A cache could return another incident's summary. Now only identical requests match





Prune's verify page: "Signature valid"

Judges attack it live: 8 tricks in the Break it round

You're on the trail now

[TO FILL: Join QR code]

1. Join
2. Zone alert
3. Crowd warning
4. Offline SOS
5. One incident
6. Are you safe?
7. Accept
8. Family link
9. Signed receipt

Demo zones and dispatches are simulated.

SafeTrail doesn't replace 112. It's the safety layer before it.

Digital Tourist ID

The pass links to the permit a traveller already holds. Built today with a simulated registry.

Where it starts

Routes that already need registration or permits: Kedarnath and the Char Dham yatra, the North East, national parks.

How it could pay

A state or park licenses it, with a small fee per tourist.

github.com/gaurvit06/Safetrail

Questions judges will ask

Question	Answer
Why not just call 112?	We warn before it happens, work across languages, keep the SOS with no signal, and give one grouped picture. We work alongside 112.
Does it work with no network?	Warnings, the pack and saving the SOS work offline. Sending needs some signal: data, SMS or a checkpoint.
Is the AI deciding emergencies?	No. Rules set priority and plain code counts people.
Isn't this surveillance?	It's opt-in per trip with a Stop button, location history is deleted after 7 days, and family see status only.
Can GPS be faked?	We flag impossible travel, replays and forged passes. We don't claim to stop all spoofing.

What's real, simulated and not built

Real

- Working app, 76 + 6 automated tests
- Kedarnath trail from OpenStreetMap, 15.1 km
- Live weather from Open-Meteo
- Live AI through Prune with Gemini
- Signed, verified receipts

Simulated, labelled

- Demo zones and dispatches
- Permit registry, with fake numbers
- CSED block route: help points are placeholders
- Sikkim and Corbett routes, approximate

Not built, on purpose

- Real 112 integration
- Real permit APIs
- Face recognition
- Satellite SOS
- A chatbot

Benchmark: 100 simulated trips, four ways

Run	Tokens per trip	Paise per trip	Paise per emergency	Calls to Google	From cache	Standard text
Naive (a low estimate)	1,997	15.9	n/a	124	0%	41 of 161
SafeTrail design	369	2.9	9.6	56	0%	0 of 56
SafeTrail + Prune, first day	403	3.1	10.0	57	1.7%	0 of 56
SafeTrail + Prune, repeat day	0	0	6.3	16	72.4%	0 of 56

- Cost is Google's token counts times its list price: $0.25 in, $1.50 out per million tokens, ₹88 to the dollar
- Every run used the small model, because the free key ran out of large model quota
- Google limits stopped 41 of the naive run's jobs, so its true use is higher and its emergency cost isn't comparable
- Prune rerouted no calls, so using the small model first is our own design
- Counts were 100% correct in all four runs, because plain code computes them
- Injection: 2 of 4 test notes were flagged, the English ones